EECS 498/CSE 598: Zero-Knowledge Proofs
Winter 2026
Lecture 7

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Agenda for this lecture

- Announcements
- Analyzing sumcheck
- Polynomial commitments (PCs)
- A PC with linear-sized proofs
- Interactive oracle proofs (IOPs) and *polynomial* IOPs

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Announcements

- Project 2 is online
- Autograder is working
- Due date: 2/20

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Analyzing sumcheck

Theorem 1. *Let g be an ℓ-variate polynomial of maximal degree d over the field $\mathbb{F}$. Then for any malicious prover $\mathbb{P}^*$ and value $H \neq \sum_{i \in \{0,1\}^\ell} g(i)$,*

$$\Pr[\mathsf{out}_{\mathbb{V}}[\mathbb{P}^*(g, H) \leftrightarrow \mathbb{V}(g, H)] = 1] \leq \frac{\ell d}{|\mathbb{F}|}$$

Proof:

$\mathbb{P}^*$ sends g_1 to $\mathbb{V}$

$g_1^*(0) + g_1^*(1) = H$

$r_1 \leftarrow \$ \mathbb{Z}_p$

$\xleftarrow{\quad r_1 \quad}$

$\xrightarrow{\quad g_2 \quad}$

$g_1(r_1) = g_2(0) + g_2(1)$

True w.p. $\leq \frac{d}{|\mathbb{F}|}$

$g_\ell(0) + g_\ell(1) = g_{\ell-1}(r_{\ell-1})$

$g(r_1, \ldots, r_\ell) = g_\ell(r_\ell)$

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"Regular" commitments

– lockbox

$\mathrm{Setup}(1^\lambda) \to pp$

$\mathrm{Commit}(pp, x) \xrightarrow{\$} c$

$\mathrm{Commit}(pp, x; r)$

$\mathrm{Open}(pp, c, x, r) \to \{0,1\}$

$\underline{\mathrm{ComHide}_b(\mathcal{A})}$:

$pp \leftarrow_\$ \mathrm{Setup}$

$m_0, m_1 \leftarrow_\$ \mathcal{A}(pp)$

$c_b \leftarrow_\$ \mathrm{Com}(pp, m_b)$

$b' \leftarrow_\$ \mathcal{A}(pp, c_b)$

$\underline{\mathrm{ComBind}(\mathcal{A})}$:

$pp \leftarrow_\$ \mathrm{Setup}$

$(x_0, r_0), (x_1, r_1), c \leftarrow \mathcal{A}(pp)$

$\mathrm{Open}(pp, c, x_0, r_0) = 1 \land (x_0, r_0) \neq (x_1, r_1)$

$\land\, \mathrm{Open}(pp, c, x_1, r_1) = 1$

Pedersen commitments

Setup: $G, H \leftarrow\$ \mathbb{G}$ $\qquad G_1, G_2, \ldots, G_n$

Commit($pp = (G, H)$, $x \in \mathbb{Z}_p$)
$r \leftarrow\$ \mathbb{Z}_p$
Ret $C := xG + rH$

$\Longrightarrow$ works for vectors!

$r \leftarrow\$ \mathbb{Z}_p$
Ret $(\sum x_i G_i) + rH$
vector Pedersen commitment

Open(pp, C, x, r)
as expected

Thm: Pedersen comms are hiding and binding.

Proof: exercise

Pedersen comms are additively homomorphic

Polynomial commitments

Setup(n) : pp

n → Degree bound

Commit(pp, p) :
output C_p and opening r

open(pp, C_p, p, z, r) output π
and v s.t. $p(z)=v$

verify(pp, C_p, z, v, π) : output 0/1

Interactive syntax → open[$\mathbb{P}(pp, C_p, p, z, r) \longleftrightarrow \mathbb{V}(pp, C_p, z)$]

$\mathbb{V}$ outputs 0/1

PC security

EvalBind_{PC}^{A}:

$pp \leftarrow\$ \text{Setup}(n)$

$p, z, C_p, (\pi_1, v_1), (\pi_2, v_2) \leftarrow\$ A(pp)$

Return $\text{Ver}(pp, C_p, z, v_1, \pi_1)$
$\wedge \text{Ver}(pp, C_p, z, v_2, \pi_2)$
$\wedge \; v_1 \neq v_2$

Interactive PC is sound if $\text{Open}[P \leftrightarrow V]$ is a sound IP for R_{PC}

Note: Hiding + many other soundness notions!

$$R_{PC}: \left\{ (pp, C_p, z, v, n \,;\, p, r) : \begin{array}{l} pp \leftarrow\$ \text{Setup} \\ C_p \leftarrow \text{Com}(pp, p; r) \\ p(z) = v \\ \text{and } \deg(p) \leq n \end{array} \right\}$$

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A PC with linear-sized proofs

Setup(n):

$H, G_0, \dots, G_n \leftarrow\$ \mathbb{G}$

Commit$(pp := (G_0, \dots, G_n, H)$ $p)$:

$r \leftarrow\$ \mathbb{Z}_p$

Parse p $a_0, \dots, a_n$

Ret $rH + \sum a_i G_i$

Opening protocol

$\mathsf{Open}[\mathcal{P}](\mathsf{pp}, C_{\vec{p}}, \vec{p}, n, z, v, r_p)$		$\mathsf{Open}[\mathcal{V}](\mathsf{pp}, C_{\vec{p}}, n, z, v)$
$\vec{y} \leftarrow \langle 1, z, z^2, \dots, z^n \rangle$		$\vec{y} \leftarrow \langle 1, z, z^2, \dots, z^n \rangle$
$\vec{d} \leftarrow\!\!\$\ \mathbb{Z}_p^{n+1}$; $r_1 \leftarrow\!\!\$\ \mathbb{Z}_p$		
$C_1 \leftarrow \mathsf{Com}(\mathsf{pp}, \vec{d}; r_1)$; $v_{\vec{d}} \leftarrow \langle \vec{d}, y \rangle$		
	$\xrightarrow{C_1, v_{\vec{d}}}$	
		$e \leftarrow\!\!\$\ \mathbb{Z}_p$
	$\xleftarrow{e}$	
$\vec{p}' \leftarrow e\vec{p} + \vec{d}$		
$r_{\vec{p}'} \leftarrow er_p + r_1$		
	$\xrightarrow{\vec{p}', r_{\vec{p}'}}$	
		Return $eC_{\vec{p}} + C_1 = \mathsf{Com}(\mathsf{pp}, \vec{p}'; r_{\vec{p}'})$
		$\wedge\ ev + v_{\vec{d}} = \langle \vec{p}', \vec{y} \rangle$

$$ev + v_{\vec{d}} = \langle \vec{p}', \vec{y} \rangle$$

$$e\langle \vec{p}, \vec{y} \rangle + \langle \vec{d}, \vec{y} \rangle = \langle \vec{p}', \vec{y} \rangle$$

$$\langle e\vec{p} + \vec{d}, \vec{y} \rangle = \langle \vec{p}', \vec{y} \rangle$$

$$eC_{\vec{p}} + C_1$$

$$= a_0 e G_0 + \dots + a_n e G_n + r_p e H$$

$$d_0 G_0 + \dots + d_n G_n + r_1 H$$

$$= (a_0 e + d_0) G_0 + \dots + (a_n e + d_n) G_n + r_p e + r_1 H$$

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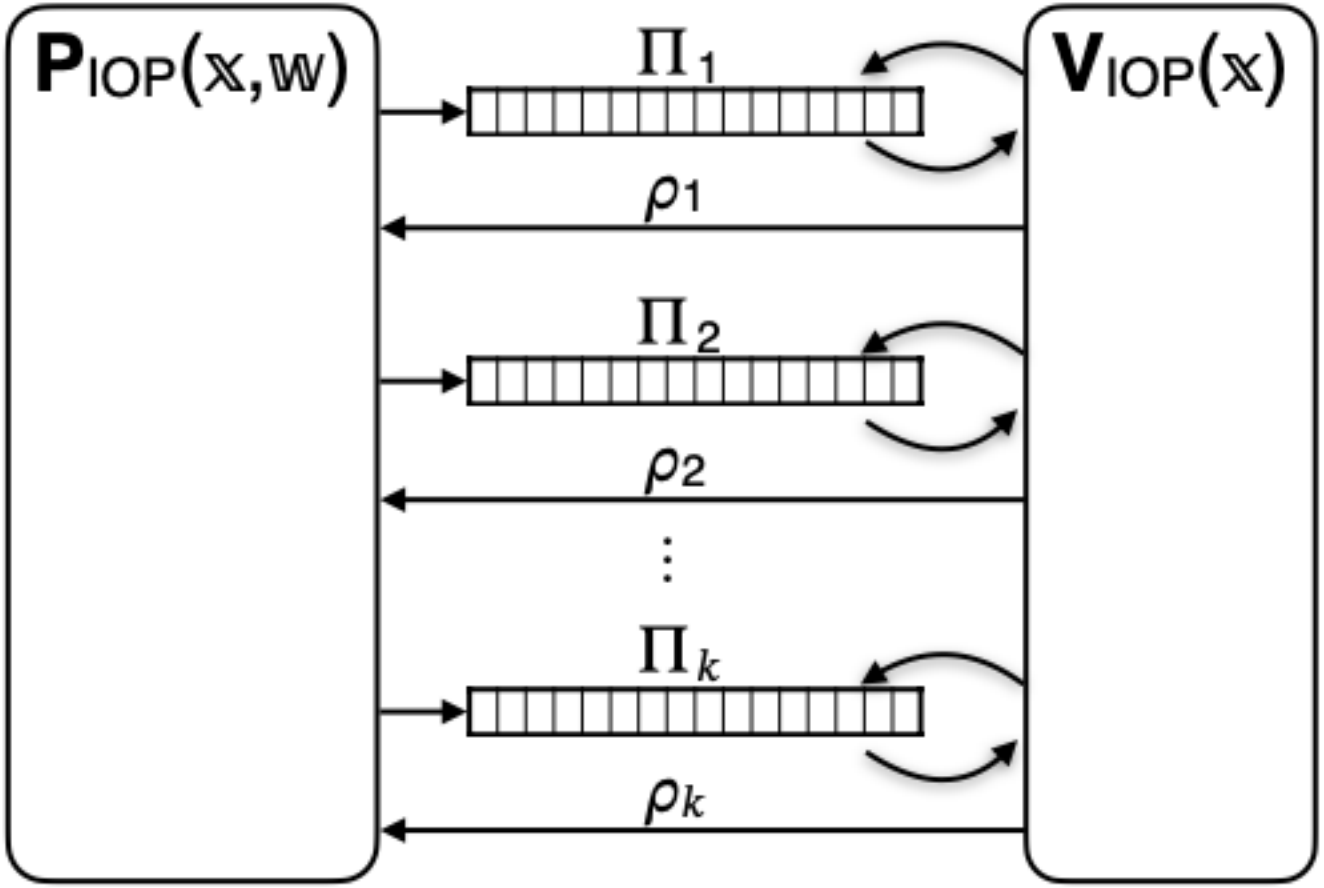
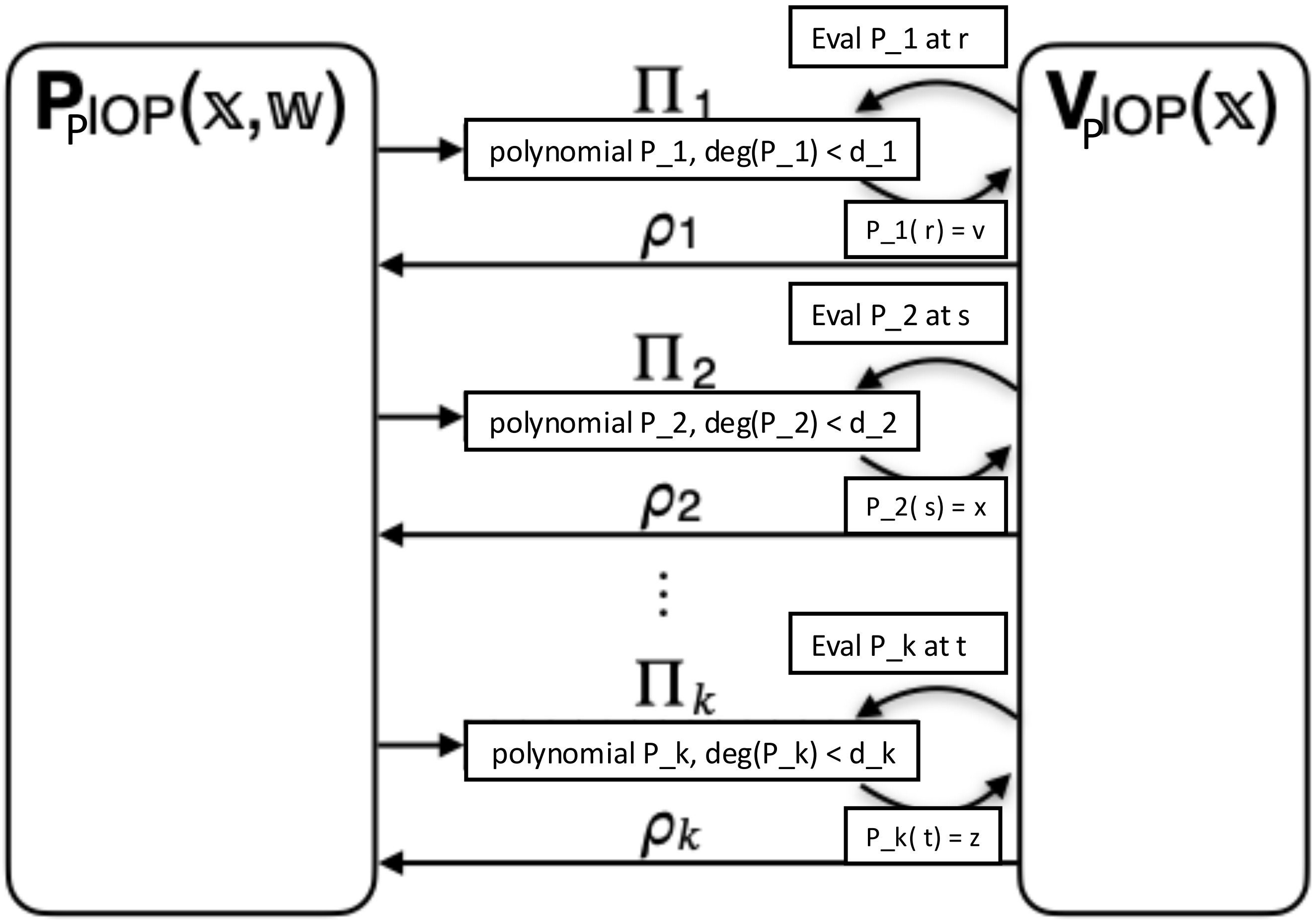
Interactive oracle proofs

Interactive oracle proofs



(Image from Chiesa-Yogev)

Polynomial IOPs



Sumcheck as a PIOP

Sumcheck.$\mathcal{P}(g, H)$		Sumcheck.$\mathcal{V}^{\mathcal{O}_g}(H)$
Compute $g_1(X_1)$ as $\sum_{x_2,\ldots,x_v \in \{0,1\}} g(X_1, x_2, \ldots, x_v)$		
	$\xrightarrow{\mathcal{O}_{g_1}}$	
		Check $H = \mathcal{O}_{g_1}(0) + \mathcal{O}_{g_1}(1)$ by querying $\mathcal{O}_{g_1}$ $r_1 \leftarrow\!\!\$\ \mathbb{F}$
	$\xleftarrow{r_1}$	
	...... In round j:	
Compute $g_j(X_j)$ as $\sum_{x_{j+1},\ldots,x_v \in \{0,1\}} g(r_1, \ldots, r_{j-1}, X_j, x_{j+1}, \ldots, x_v)$		
	$\xrightarrow{\mathcal{O}_{g_j}}$	
		Query $\mathcal{O}_{g_{j-1}}$ and $\mathcal{O}_{g_j}$ to check $\mathcal{O}_{g_{j-1}}(r_{j-1}) = \mathcal{O}_{g_j}(0) + \mathcal{O}_{g_j}(1)$ $r_j \leftarrow\!\!\$\ \mathbb{F}$
	$\xleftarrow{r_j}$	
	...... Final round:	
$g_v \leftarrow g(r_1, \ldots, r_{v-1}, X_v)$		
	$\xrightarrow{\mathcal{O}_{g_v}}$	
		Query $\mathcal{O}_{g_{v-1}}$ and $\mathcal{O}_{g_v}$ to check $\mathcal{O}_{g_{v-1}}(r_{v-1}) = \mathcal{O}_{g_v}(0) + \mathcal{O}_{g_v}(1)$ $r_v \leftarrow\!\!\$\ \mathbb{F}$ Query/check $\mathcal{O}_{g_v}(r_v) = \mathcal{O}_g(r_1, \ldots, r_v)$ If all checks pass: Return true Else return false

Sumcheck as a PIOP

Sumcheck.$\mathcal{P}(g, H)$		Sumcheck.$\mathcal{V}^{\mathcal{O}_g}(H)$
Compute $g_1(X_1)$ as $\sum_{x_2,\ldots,x_v \in \{0,1\}} g(X_1, x_2, \ldots, x_v)$		
	$\xrightarrow{\mathcal{O}_{g_1}}$	
		Check $H = \mathcal{O}_{g_1}(0) + \mathcal{O}_{g_1}(1)$ by querying $\mathcal{O}_{g_1}$ $r_1 \leftarrow\$ \mathbb{F}$
	$\xleftarrow{r_1}$	
	In round j:	
Compute $g_j(X_j)$ as $\sum_{x_{j+1},\ldots,x_v \in \{0,1\}} g(r_1, \ldots, r_{j-1}, X_j, x_{j+1}, \ldots, x_v)$		
	$\xrightarrow{\mathcal{O}_{g_j}}$	
		Query $\mathcal{O}_{g_{j-1}}$ and $\mathcal{O}_{g_j}$ to check $\mathcal{O}_{g_{j-1}}(r_{j-1}) = \mathcal{O}_{g_j}(0) + \mathcal{O}_{g_j}(1)$ $r_j \leftarrow\$ \mathbb{F}$
	$\xleftarrow{r_j}$	
	Final round:	
$g_v \leftarrow g(r_1, \ldots, r_{v-1}, X_v)$		
	$\xrightarrow{\mathcal{O}_{g_v}}$	
		Query $\mathcal{O}_{g_{v-1}}$ and $\mathcal{O}_{g_v}$ to check $\mathcal{O}_{g_{v-1}}(r_{v-1}) = \mathcal{O}_{g_v}(0) + \mathcal{O}_{g_v}(1)$ $r_v \leftarrow\$ \mathbb{F}$ Query/check $\mathcal{O}_{g_v}(r_v) = \mathcal{O}_g(r_1, \ldots, r_v)$ If all checks pass: Return true Else return false

Pop quiz: what is the degree bound for the g_i oracles sent during the proof?